

Elementary Operations on Quasi-Toeplitz Matrices

PCCA6

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- Operations on large structured matrices are often limited by memory bandwidth.
- Standard operations on n^2 elements lead to unnecessary computations.

Goal

Optimize memory footprint and execution time on structured matrices by implementing displacement operators and generator-based representations.

Where do we see these structured matrices?

- Sylvester matrix
- Diagonal gradients
- Quantum channel calculations in classical quantum simulations
- ...

Definition

A matrix $A \in \mathbb{K}^{n \times m}$ is **Toeplitz** if $A_{i,j} = A_{i+1,j+1}$ for all valid i, j .

$$A = \begin{pmatrix} a_0 & a_{-1} & \cdots & a_{-(m-1)} \\ a_1 & a_0 & \cdots & a_{-(m-2)} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n-1} & \cdots & a_1 & a_{n-m} \end{pmatrix}$$

Displacement & Generators

Displacement Operator

$$\nabla A = A - ZAZ^T = GH^T$$

For Toeplitz: $\text{rank}(\nabla A) \leq 2$

Notice

Storing (G, H) instead of A reduces memory from $O(n^2)$ to $O(\alpha n)$.

Reconstruction

$$A = \sum_{k=1}^{\alpha} L(g_k) U(h_k)$$

Example - Displacement & Generators

Toeplitz Matrix A

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix}$$

Displacement $\nabla A = A - ZAZ^T$

$$\nabla A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 \end{pmatrix}$$

Only first row and column stays. $\text{rank} = 2$.

Addition of Toeplitz-like Matrices

Generator Addition

$$A = (T, U), B = (G, H) \implies A+B = (T|G, U|H)$$

Notice

from $O(n^2)$ to $O(\alpha M(n))$.

Multiplication of Toeplitz-like Matrices

Generator Multiplication with Vectors

blablabla

Notice

from $O(???)$ to $O(???)$.

Multiplication of Toeplitz-like Matrices

Generator Multiplication

$$A = (T, U), \quad B = (G, H)$$

$$W = ZAZ^t G$$

$$V = B^t U$$

$$\alpha = ZA$$

$$\beta = ZB^t$$

$$C = (T|W|\alpha, V|H|\beta)$$

Notice

from $O(n^3)$ to $O(\alpha^2 M(n))$.

Thank you